

Welcome to Code Crunch!

C5 | CRUNCH LABS: AI & Data Science

[Mon 1-2PM, Wed 12:30-1:30PM]
Weekly | Spring 2025

Attendance

Earn a Certificate of
Completion by attending
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Unit #7

Neural Networks

C5|Crunch Labs: Crunch AI Data Science



Neural Network

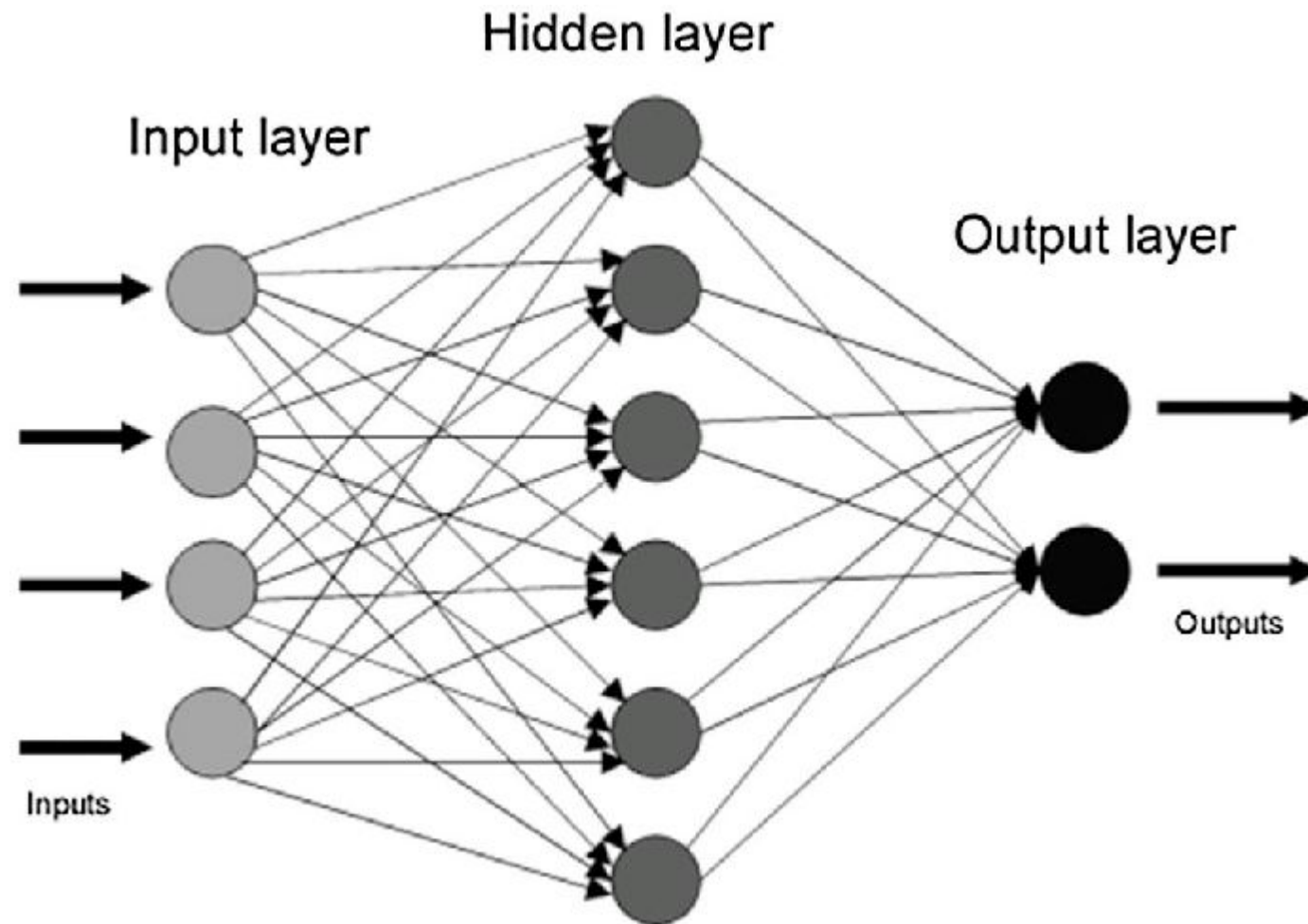
What are Neural Networks?

- Neural networks are a type of machine learning model inspired by the structure and functioning of the human brain.
 - They are designed to recognize patterns and make predictions or classifications based on data.

Why are neural network layers?

- A neural network consists of layers of interconnected nodes, also known as neurons. These layers are organized into three main types.
 - **Input layer:** This is where the data is fed into the network. Each neuron in this layer represents a feature or attribute of the data.
 - **Hidden layers:** These layers are where most of the computation happens.
 - Neurons in these layers process the input data, applying mathematical transformations to extract patterns or features.
 - There can be multiple hidden layers, which helps the network learn complex relationships.
 - **Output layer:** The final layer produces the result of the network's computation, such as a prediction or classification.

cont.





how do they work?

Neural Networks weights?

- The neurons in each layer are connected by weights.
 - These weights determine the strength of the connection between neurons and are adjusted during the training process to minimize errors in predictions.
 - The network learns by backpropagation, where the error is calculated and the weights are updated accordingly.

what are some applications?

- Neural networks are commonly used in a variety of applications, including:
 - image recognition
 - natural language processing
 - speech recognition
 - autonomous driving.
- The most complex neural networks are deep neural networks (DNNs), where there are many hidden layers—leading to the term deep learning.



Neural Network layers

Convolutional Layer (spatial data)

- Used in convolutional neural networks (CNNs).
- Extracts spatial features from input data (e.g., images).
- Uses filters (kernels) to detect edges, textures, and patterns.

Pooling Layer (Sequential data)

- Used in recurrent neural networks (RNNs) for sequential data (e.g., time series, text).
- Stores memory of previous inputs.

Dropout Layer (data)

- Stores memory of previous inputs.
- Reduces overfitting by randomly deactivating neurons during training.

Flatten Layer (spatial data)

- Converts multi-dimensional data into a one-dimensional vector (often used before fully connected layers).

Embedding Layer (sequential/non numerical data)

- Used in NLP tasks to map categorical data (words) into dense vector representations.

cont.



Attention Layer (sequential - spatial)

- Helps models focus on important parts of input sequences.
- Used in transformers and NLP tasks.

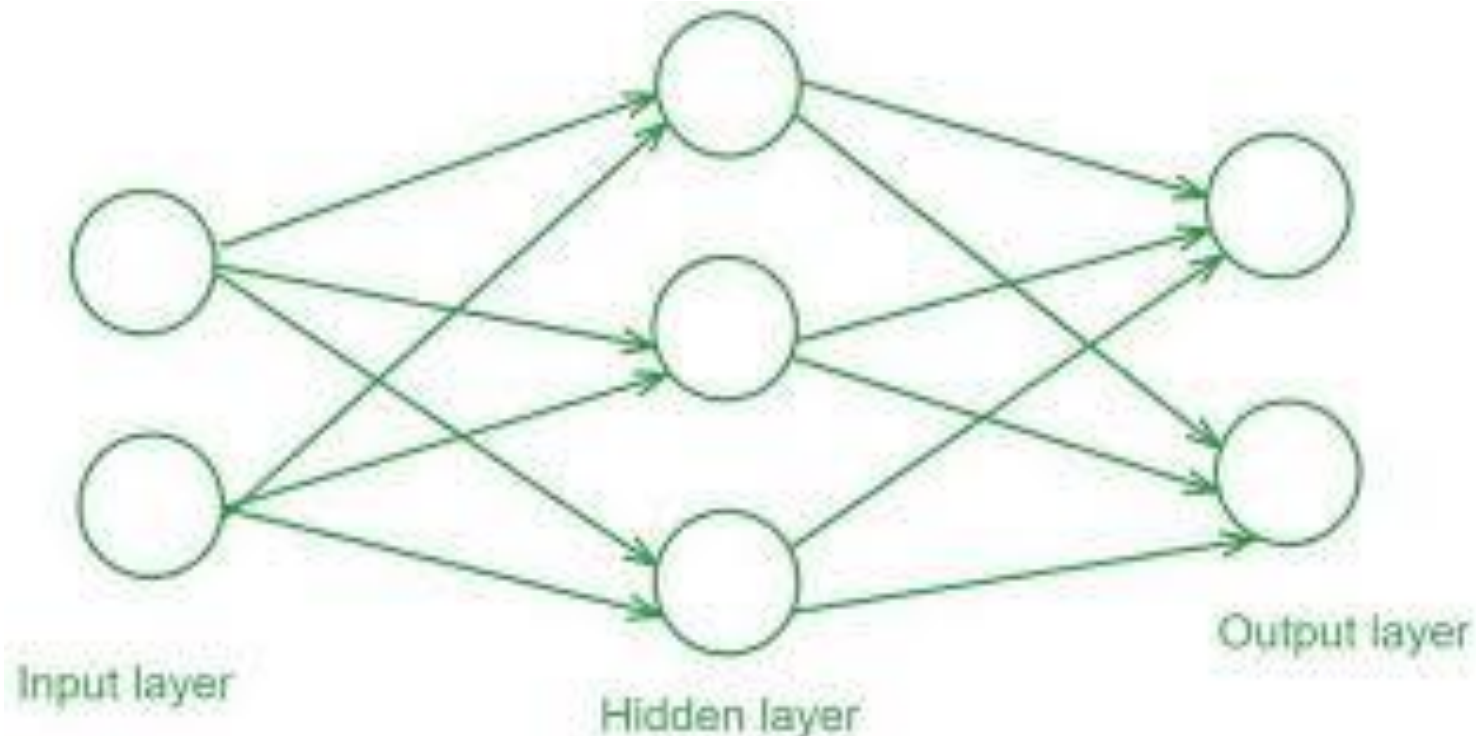
Softmax Layer (data)

- It is commonly used in the final layer of a classification neural network to convert raw scores (logits) into probabilities.

Types of Neural Networks

1. Feedforward Neural Networks (FNN)

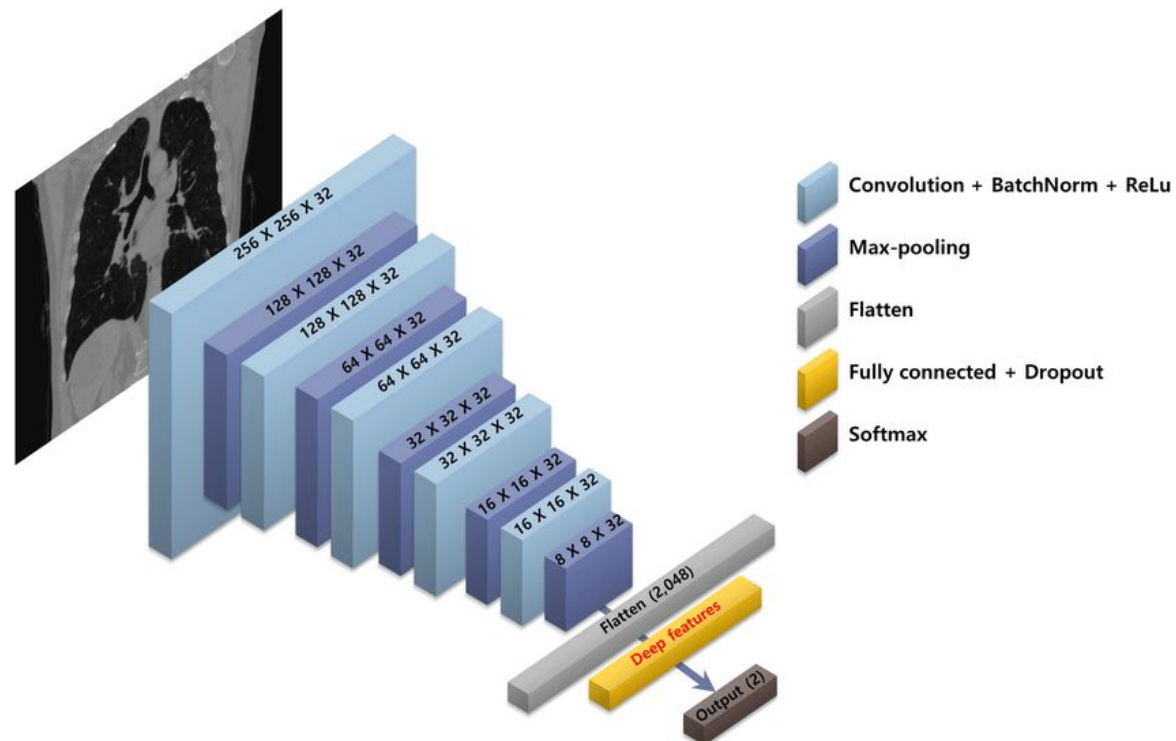
- The simplest type of neural network.
- Information moves in one direction—from input to output—without loops or cycles.
- Used in pattern recognition and simple classification tasks.



Types of Neural Networks

2. Convolutional Neural Networks (CNN)

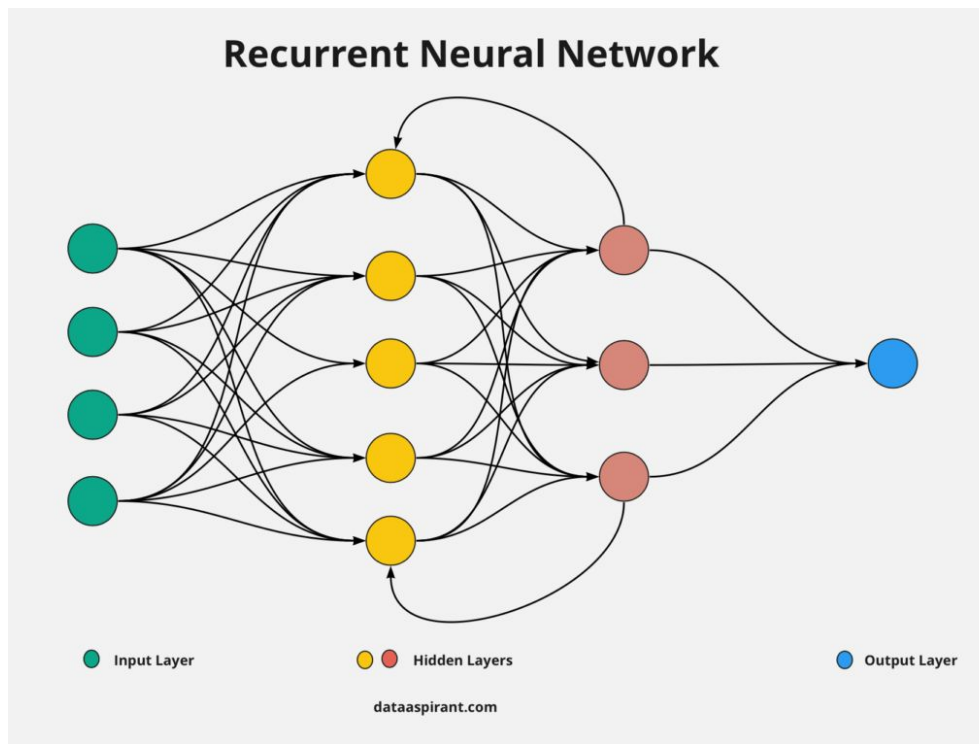
- Primarily used for image and video processing.
- Uses convolutional layers to detect spatial hierarchies in data.
- Applications: Image recognition, object detection, facial recognition.



Types of Neural Networks

3. Recurrent Neural Networks (RNN)

- Designed for sequential data, where past information influences the output.
- Uses loops to store information from previous inputs.
- Applications: Speech recognition, language modeling, time-series forecasting.



Types of Neural Networks

4. Transformer Networks

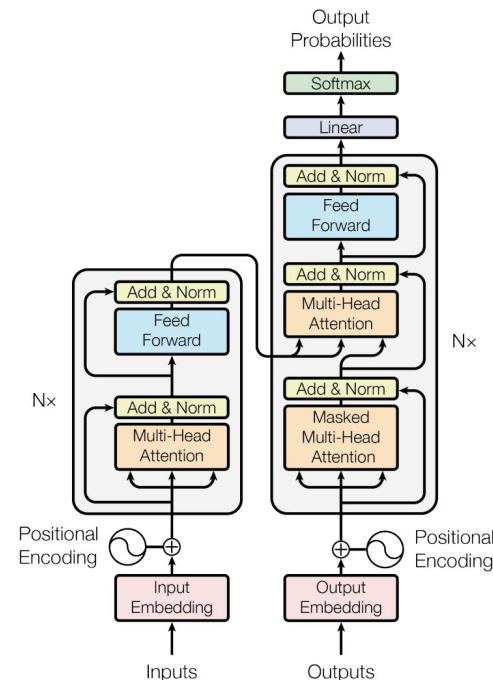
- Uses self-attention mechanisms to process sequential data efficiently.
- Forms the basis of models like GPT and BERT.
- Applications: Natural language processing (NLP), machine translation.

BERT

Encoder

GPT

Decoder





Q&A Session



Attendance



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Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



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Go to our website: go.fiu.edu/codecrunch